Preconditions

The types of the supplied lockable objects `LockableType1, LockableType2`,... shall conform to the `Lockable` requirements.

Effects

Tries to acquires a lock on each of the supplied lockable objects `m1, m2`,... by calling `try_lock()` on each in turn. If a call to `try_lock()` returns `false` or throws an exception, locks already acquired are released by calling `unlock()` on the corresponding lockable object.

Returns

`-1` if all locks were acquired (each call to `try_lock()` returned `true`), otherwise the zero-based index of the object for which the call to `try_lock()` returned `false`.

Postconditions

If the function returns `-1`, the current thread owns a lock on each of the supplied lockable objects. Otherwise, any locks acquired by this call have been released.

Throws

Any exceptions thrown by the calls to `try_lock()`.

Note

If an exception propagates out of the call to `std::try_lock`, then `unlock()` shall have been called for any of the objects `m1, m2`,... for which a lock has been acquired in the function by a call to `try_lock()`.
